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Statement of Purpose

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Modern software development is a collaborative effort where various teams contribute artifacts like source code, interfaces, multimedia content, and documentation. However, researchers mainly focus on source code, overlooking other artifacts [1]. One such artifact is build systems, which orchestrate the transformation of source code into deliverables. A poorly maintained build system can lead to frequent breakages, thus creating barriers to efficient software engineering (SE) [2]. Motivated by these challenges, I want to address these barriers during my Ph.D. My goal is to first investigate inefficiencies that waste computing resources and hinder developer productivity. Then, propose automated solutions around developers' behaviors and expectations to mitigate these inefficiencies and accelerate build jobs.

Throughout my master's research in SE, I have recognized that many automated solutions can go underutilized in practice. For example, in a graduate course, I studied developers' preferences for various settings of program repair tools and found a notable misalignment. That is, researchers have built a plethora of repair tools that minimize developer involvement. However, developers prefer tools that offer them some control in the issue resolution process, as also discussed in prior work [4]. Additionally, many developers were unaware of these tools and most times turned to ChatGPT for help [3]. This experience has made me realize that significant advancement in SE research can be overlooked unless the automated solutions are designed with developers' behaviors and expectations in mind.

Therefore, I am currently collaborating with Professor X to explore how developers use ChatGPT for issue resolution, with the aim of providing practical solutions. To this end, we conducted a mixed-method empirical study to investigate patterns in developers' use of ChatGPT. Our findings indicate that developers primarily utilize ChatGPT for brainstorming solutions for issues, while AI-generated code was directly applied to resolve 12.78% of those issues. Additionally, our study suggests that more context-dependent tasks, such as debugging, receive less assistance from ChatGPT. This indicates the need for a tool that can automatically retrieve context for these tasks. The work was published at SANER, received a distinguished paper award, and was invited for a journal extension in EMSE.

All these prior experiences have inspired me to pursue a Ph.D in SE with the vision of proposing tools that meet developer needs. I am particularly interested in Professor X's work on tools and decision support for modern software development and release teams, as it closely aligns with my research objectives. This has been confirmed by my conversations with the professor, as well as discussions about potential collaborations with Professor X's Ph.D. student. Lastly, UWaterloo is an excellent choice for pursuing my Ph.D, primarily due to its strong SE research and faculty.

References

[1] Robles, G., et al. "Beyond source code: The importance of other artifacts in software development." JSS (2006).

[2] Nejati, M., et al. "Code review of build system specifications: Prevalence, purposes, patterns, and perceptions." ICSE (2023).

[3] Meem, FN., et al. "Exploring experiences with automated program repair in practice." ICSE (2024).

[4] Winter, E., et al. "How do developers really feel about bug fixing? directions for automatic program repair." IEEE TSE (2022).